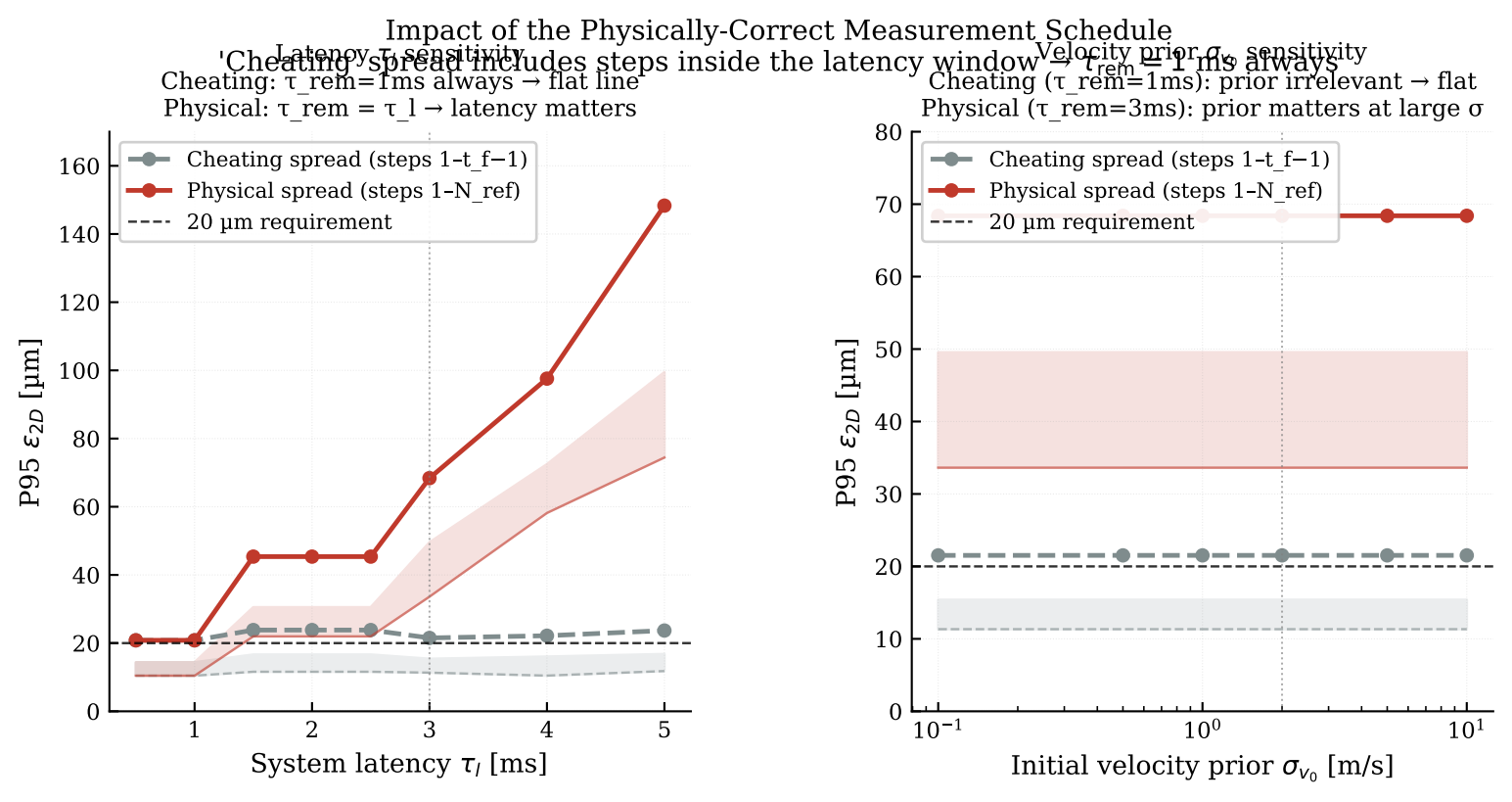
Impact of the Physically-Correct Measurement Schedule
'Cheating' spread includes steps inside the latency window → $\tau_{rem} = 1$ ms always
Latency τ_l sensitivity
Cheating: τ_rem=1ms always → flat line
Physical: τ_rem = τ_l → latency matters
Cheating spread (steps 1–t_f−1)
Physical spread (steps 1–N_ref)
20 µm requirement
P95 ε_{2D} [µm]
System latency τ_l [ms]
Velocity prior σ_{v_0} sensitivity
Cheating (τ_rem=1ms): prior irrelevant → flat
Physical (τ_rem=3ms): prior matters at large σ
Cheating spread (steps 1–t_f−1)
Physical spread (steps 1–N_ref)
20 µm requirement
P95 ε_{2D} [µm]
Initial velocity prior σ_{v_0} [m/s]